

Recommendation for the 2026–2027 Schedule

RANGER class • MATE ROV Competition

Twelve months. Mid-June 2026 to Dauphin Island.

Strategy

A new RANGER team will not out-build experienced teams in ten months. But it can out-prepare them on documentation, presentation, and safety — together, forty percent of the total score. So this plan front-loads skills, fundamentals, and the company's identity over the summer and fall, and treats the spring as execution against a known manual. Everything is in service of one deliverable, late April 2027: Dauphin Island.

The plan

Phase one — Foundation. Mid-June through July 2026.

Six weeks. The goal is not a competition ROV. It is a team that exists, a working starter ROV in the water, and kids who have soldered something. Form the company — CEO, CFO, mechanical lead, electrical lead, software lead, communications lead. Build the SeaMATE Triggerfish kit as the team's first ROV — a well-documented, MATE-blessed training platform that gets every kid hands-on with the same components they'll later design from scratch. Get every kid in a pool by week three. Lock pool access for the year, including one deep-water host. Hold a soldering bootcamp, an electronics workshop, and a project-notebook session. Add a waterproofing exercise: each kid (or pair) designs a small sealed PVC capsule with a single cable penetration, drops it to 2 m, recovers it, checks the paper towel inside, and iterates until it stays dry across three consecutive dives — the foundation skill for the float build in Phase 2. Start the community-outreach log on day one — every hour becomes Corporate Responsibility points in April.

SeaMATE Triggerfish ROV kit	\$950
Two soldering stations and consumables	\$130
Two multimeters and hand tools	\$110
Waterproofing test-capsule exercise (PVC, end caps, O-rings, epoxy)	\$50
<i>Subtotal</i>	\$1,240

Phase two — Capability. August through mid-October 2026.

Ten weeks. School starts. Settle into a weekly cadence: one weeknight, one weekend block. Each sub-team takes on a mini-project that maps to a real RANGER skill. Mechanical designs and 3D-prints a working gripper. Electrical builds a fused topside control box with proper

terminations. Software writes a Python program that detects and counts objects in a webcam feed. The float team builds a v1 vertical profiler with a syringe buoyancy engine that hits 2.5 m and transmits depth over LoRa — this is now an application of the waterproofing skill learned in Phase 1, not a first encounter with it. Every member completes the Onshape learning path and captures a household object with photogrammetry. The prototypes are throwaway. The learning is the deliverable.

Float v1 components (microcontroller, MS5837 sensor, LoRa pair, syringe, battery)	\$200
3D-printer filament (community maker-space access assumed)	\$50
Gripper and payload prototype materials	\$60
Topside control prototype (Raspberry Pi 4, gamepad, breadboard)	\$120
USB camera for computer-vision work	\$30
Outreach materials and workshop consumables	\$125
<i>Subtotal</i>	\$585

Phase three — Design. Mid-October through mid-January 2027.

Twelve weeks. The 2027 manual drops around November. Within forty-eight hours of release, the team reads it together. Then a two-week decomposition: every task broken into requirements, scoring, and constraints. Trade studies follow — frame architecture, thruster count, camera choices, payload tools, float comms protocol, buoyancy-engine type. Every decision earns a written rationale in the project notebook; those rationales become the technical documentation in March. CAD locks by mid-December. Long-lead parts — thrusters, tubes, penetrators — ship before the holiday break. The break is the natural pause.

Thrusters: six Rule 500 GPH bilge pumps, propellers, couplers	\$260
Motor drivers (BTS7960 H-bridges, six channels)	\$60
Watertight enclosure (PVC tube, acrylic ends, epoxy, O-rings)	\$90
Two underwater cameras (USB webcams in DIY housings)	\$80
WetLink penetrators, eight at \$13	\$105
Tether: 100 ft Cat5e signal cable plus integrated power conductors	\$150
Raspberry Pi 5 (ROV main) and Arduino Mega (sensor interface)	\$170
Frame materials (HDPE or PVC plus fittings)	\$80

Topside 12 V power supply	\$50
Float v2 refinement parts	\$80
Contingency buffer	\$100
<i>Subtotal</i>	\$1,225

Note on thruster choice: Blue Robotics T200 thrusters cost about \$230 each — six of them alone would consume \$1,380. Bilge-pump conversions are the standard low-budget RANGER approach and produce competitive results when paired with good control software.

Phase four — Build. Mid-January through February 2027.

Six weeks. Construction. Tether, topside control box, ROV frame, electronics integration, float build, payload tools. Plan for two weeks of “it doesn’t work” before it works. Bare frame wet by end of January. Electronics integrated by mid-February. First full-system test by end of February. The fluid-power quiz and safety pre-inspection tracking begin now.

Rework consumables (epoxy, sealants, O-rings)	\$80
Mounting hardware, fasteners, brackets	\$80
Replacement-parts buffer (one major break)	\$150
Display-board base materials	\$60
Misc build consumables	\$50
<i>Subtotal</i>	\$420

Phase five — Test and document. March 2027.

Four weeks. Full mission rehearsals at the deep-water pool every weekend. Time each task. Identify what breaks; fix it. Lock the SIDs — electrical, fluid power if used, non-ROV device. Write the JSEA, focused on poolside operations rather than shop safety. Draft the technical documentation. Run two mock engineering presentations with parents acting as judges.

Critical spares (motor, driver, penetrator backups)	\$200
Marketing display printing (Staples 36 × 48) and mounting	\$130
Documentation printing and binding	\$50
Backup hardware (spare Pi, sensors)	\$80
<i>Subtotal</i>	\$460

Phase six — Polish and compete. April 2027 through the regional.

Documentation submission deadlines hit in early April. Confirm DISL’s exact dates. Print the marketing display, thirty-six by forty-eight inches. Five engineering-presentation rehearsals, minimum. Pack the trailer. Drive to Dauphin Island, late April 2027.

Regional registration fee	\$200
Final marketing-display reprint and handouts	\$60
Team shirts (eight at \$20)	\$160
Day-of miscellaneous	\$50
<i>Subtotal</i>	\$470

Skills the team will leave with

By April, every kid will have hands-on with: soldering and basic electronics, fusing and full-load-amp measurement, waterproofing — O-rings, epoxy potting, penetrators — 3D CAD in Onshape, 3D printing, Python with either Arduino or Raspberry Pi, OpenCV for image recognition, photogrammetry, pressure and depth sensors, wireless RF, basic project management, technical writing, and public speaking with live Q&A. Specialists will go deeper in their lanes.

Budget summary

Working total: \$4,400, based on real 2026 prices for the SeaMATE kit, bilge-pump thrusters, BlueRobotics penetrators, and Staples marketing-display printing. Excludes deep-pool access (donated facility), project notebooks (kids bring their own), and travel, lodging, and meals for the competition trip — those are tracked separately through fundraising and parent contributions.

Phase one — Foundation	\$1,240
Phase two — Capability	\$585
Phase three — Design	\$1,225
Phase four — Build	\$420
Phase five — Test and document	\$460
Phase six — Polish and compete	\$470
Total	\$4,400

Milestones

End of July 2026	Starter ROV in the water. Deep-pool access confirmed. Company formed.
November 2026	2027 manual released. Full-team read-through within forty-eight hours.
Mid-December 2026	CAD locked. Long-lead parts ordered.
End of February 2027	First full-system pool test.
Early April 2027	Technical-documentation submission deadline. Confirm with DISL.
Late April 2027	Northern Gulf Coast Regional. Dauphin Island Sea Lab.

Risks, and how to defuse them

Two failure modes kill new RANGER teams. The first is not getting in the water enough. Fix this by locking pool access in June, not October. The second is under-investing in the engineering-and-communications track because the ROV isn't done yet. Fix this by starting the marketing display and the technical-documentation skeleton in August, even when there is little to put in them.

A third risk lives at the parent level. Over-functioning. The rule for the year is: teach a skill once, then step back and let them fail. The kids learning to recover from failure is what makes the engineering-presentation Q&A go well in April.

Where to start

If we do nothing else in June: buy the SeaMATE Triggerfish kit, get six kids soldering, and put it in the water within three weeks. Everything else in this plan is downstream of that.

Appendix — Integrated Master Schedule

One page. Six phases. Eleven months. The bars show duration; the dots below the timeline mark the milestones above. Pin it to a wall.

